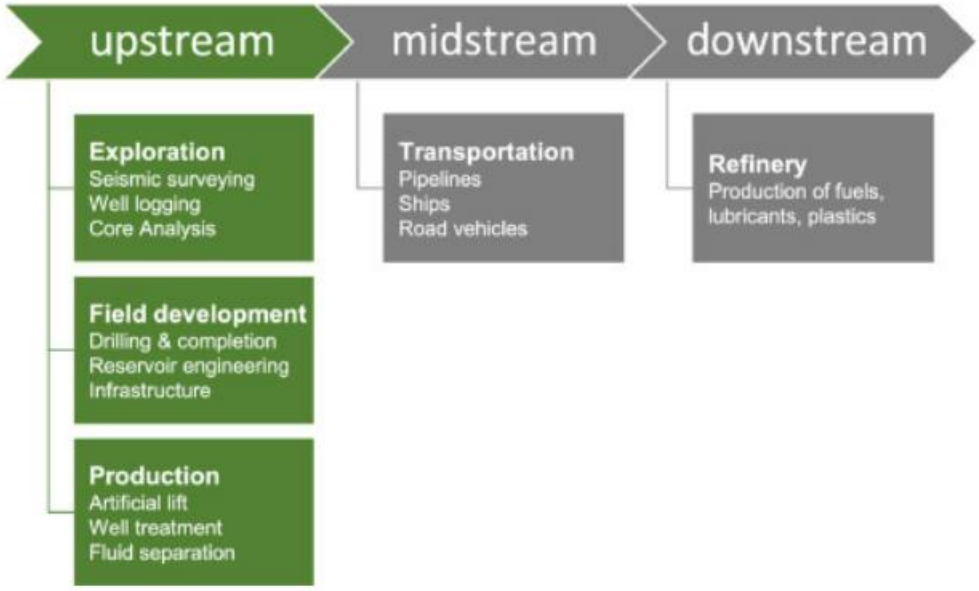
APPLICATION OF AI/ML IN OIL AND GAS INDUSTRY

Oil and gas companies must collect and study a substantial amount of data before and after drilling into the Earth. To boost efficiency in day-to-day operations, they need to be able to solve complex exploration and production problems before they end up wasting loads of money on drilling into an unproductive well.

The OGI is among the largest, complex and crucial industries in the world, with series of upstream, midstream and downstream activities taking place, including exploration, production, refining, transportation and consumption. The industry is further characterised by working and contractual relationships with governments, joint ventures and other stakeholders. It is highly dynamic in nature, making it difficult for the traditional centralised approach to work, especially with the rise in global demand. OGI organisations are considering new ways of cutting operations cost, enhancing efficiency and increasing out-turn[(2), pg2, 1.]. Challenges and risks associated with activities and operations throughout the supply chain require to be addressed profoundly, which compels the advancement of technology within the OGI, especially for data processing and management. Some examples of such technologies include digitisation of oil fields, real-time optimisation of drilling operations, the use of nanotechnology, wireless sensor networks to aid gauging, reservoir modelling and diagnostics[(2), pg2, 2.1].



[(4), pg3, fig1.]

The upstream mainly involves exploration and production of crude. The midstream extends both upstream and downstream and involves storage and transportation; however, it is often ignored when listing out the OGI supply chain phases.

The downstream is responsible for oil and gas refining, as well as distributing the end products. Each level of the supply chain is associated with complexities that require strategic, tactical and operational decision making.

Production is the first step in the OGI supply chain; it involves drilling, extraction and recovery of oil from beneath the earth. During production, crude oil is produced both on land

(onshore) and in shallow waters (offshore). To accomplish successful production of crude oil and natural gas, different equipment is utilised and various production operators are involved being responsible for organising and supervising all operations in searching for and extracting oil.[(2), pg2, 2.2]

AI tools need the good quality data of a suitable volume to be trained and then to work properly in the operational mode. While using smarter algorithms may help in getting better results from datasets of limited size, no manipulation can help with bad data. Thus, access to big and quality data is a crucial enabler and barrier for AI applications' successful development. Oil and gas fields generate large amounts of raw data. Still, it is not a guaranty for success as there are known issues with the quality and accuracy of field data and overall lack of large volumes of labelled data in the oil and gas industry. Training datasets have to be carefully collected through the well-planned workflow- and situation-specific multi-year procedure. [(4), pg7, 6.2]

MACHINE LEARNING IN OIL AND GAS INDUSTRY APPLICATIONS

In order to minimize uncertainty, the primary step is to create system that can handle several hypotheses for achieving optimized solution. ML has a great potential in OGI applications, especially when it comes to analysing and interpreting data. It offers better ways of developing drilling plans, diagnosing, monitoring, predicting and performing real-time optimisation at a minimal cost. ML algorithms that have been effectively applied to OGI tasks include Support Vector Machine (SVM), Artificial Neural Networks (ANN), Deep Learning (DL) and Genetic Algorithms (GA). Applications using these algorithms are reviewed below in turn. [(2), pg3, 3.]

AI-AIDED EXPLORATION [(4), pg3, 3.1]:

Exploration of oil and gas reserves is a set of operations resulting in a 3D geological model of an oil/gas field or reservoir. The operations include geophysical and petrophysical studies and processing of the data acquired during the studies. Geophysical and petrophysical studies typically consist of:

1) Reservoir-scale seismic surveying

2) Well logging

3) Lab core analysis and (in some very specific cases) digital core analysis.

ALGORITHMS [(1), pg380, 2.]:

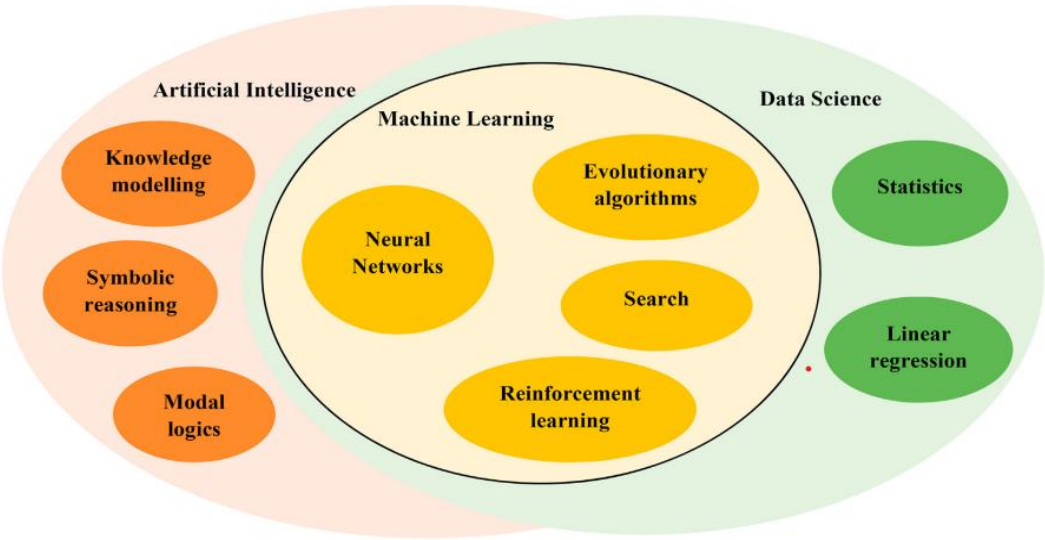
Machine Learning is a subset of Artificial Intelligence. In oil and gas industries, various types of data are collected from surface and subsurface to understand the hydrocarbon potential. The sensors are found to be most prominent to collect these data in large number. It is required to plot and analyse these data with technical analysis and intervention. The machine learning methods provides relationship between input variables and predicts the output. In machine learning, the physical behaviour of the system is not interfered. Several algorithms used are summarized below:

1.ARTIFICIAL NEURAL NETWORK (ANN) [(1), pg381, 2.1]:

ANN is used as effective machine learning method to solve complicated problems. In oil and gas industries, ANN is most widely used in nonlinear and complex problems which cannot be

solved by linear relationship. The ANN model can predict closure pressure with learning from patterns in data. The output data are generally compared with actual results to minimize error.

ANN model helps to predict pipeline conditions, it enables operators to assess and predicts the conditions of pipelines.



[(1), pg381, fig.1]

2.FUZZY LOGIC [(1), pg382, 2.2]:

Neuro Fuzzy is generally used to study well placement optimization.

3.LINEAR REGRESSION [(1), pg382, 2.4]:

The linear regression is a statistical method. There is correlation between process variables in linear regression.

4.PRINCIPAL COMPONENT ANALYSIS (PCA) [(1), pg382, 2.5]:

The principal component analysis use common patterns and trends from big data and uses it for production forecasting. Principle components methodology is generally used to forecast production from liquid rich shale reservoirs. Singular Value Decomposition (SVD) was used to calculate principal component.

MACHINE LEARNING IN UPSTREAM:

Upstream activity, tool for the application and artificial intelligence approach.

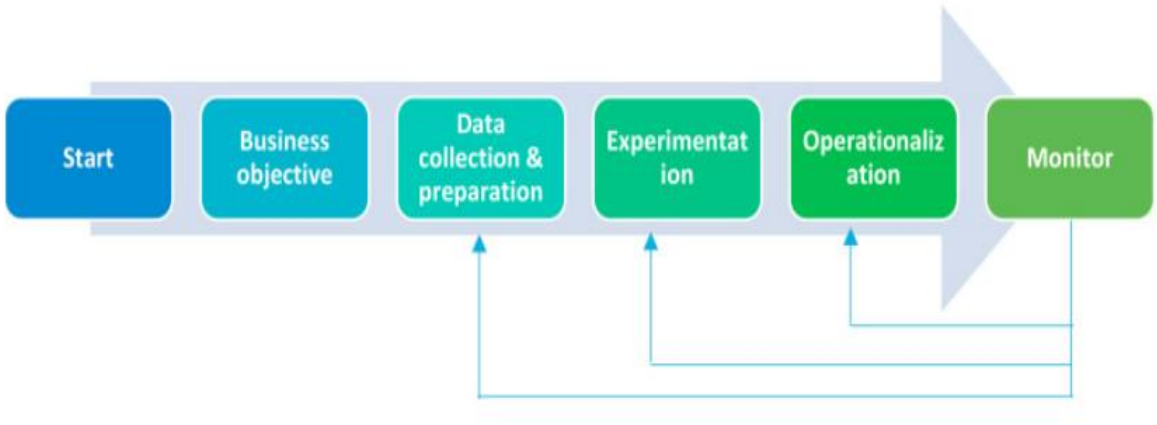
Activity	Tool for the application	Artificial Intelligence Approach
Evaluation of the subsurface geology	• A tool for automatically mapping the characteristics of reservoir rock over an oil field. • A programme for collecting geological data from well logs. Boosting the gradient by 100 times or more accelerates the process. Based on photos of rock samples collected from wells, a tool for rock typing has been created.	• Interpolation techniques + none gradient optimization • Gradient boosting • Deep neural network
Drilling	Using real-time drilling telemetry, this tool can detect the drilled rock form and possible failure.	Algorithms for machine learning in combination
Reservoir engineering	Traditional reservoir simulations can be sped up with this tool.	Deep neural networks
Production optimization	A data-driven method for predicting the efficacy of well care campaigns objectively.	Gradient boosting + feature selection based on expert opinion

[(1), pg384, Table.1]

Huge data volume is utilised to prepare reservoir maps which are refined iteratively based on new data up-gradation in database. ANN is used for estimation of reservoir properties like permeability and porosity from many years. The study can be performed by applying different machine learning methods like K Nearest Neighbours (KNN), Support Vector Regression (SVR), Kernel Ridge Regression (KRR), Adaptive Boosting and Collaborative Filtering to predict reservoir fluid properties. [(1), pg385, 3.2]

Upstream activity	Developed tool	AI approach
Geological assessment	Tool for automated mapping of reservoir rock properties over an oil region	None gradient optimization + interpolation techniques
	Tool for extracting the geological information from well logs	Gradient boosting
	Tool for rock typing based on images of rock samples extracted from the wells	Deep neural networks
Drilling	Tool for detecting the drilled rock type and potential failure using real-time drilling telemetry	Combination of machine learning algorithms
Reservoir engineering	Tool for accelerating the conventional reservoir simulations	Deep neural networks
Production optimization	Data-driven tool for an objective forecast of efficiency of well treatment campaigns	Gradient boosting + expert based feature selection

Continuous Integration/Continuous Deployment (CICD) practices in ML are yet another important application in oil and gas industry. Advanced CICD should include an accurate and reproducible Machine Learning (ML) pipeline with the mechanisms for tracking, model lineage and version control. This is especially helpful in acknowledging conceptual drift where the performance of a statistical model deteriorates over time due to changes in data and input-output relationships modelled previously. [(1), pg385, 3.2]



[(1), pg386, fig7.]

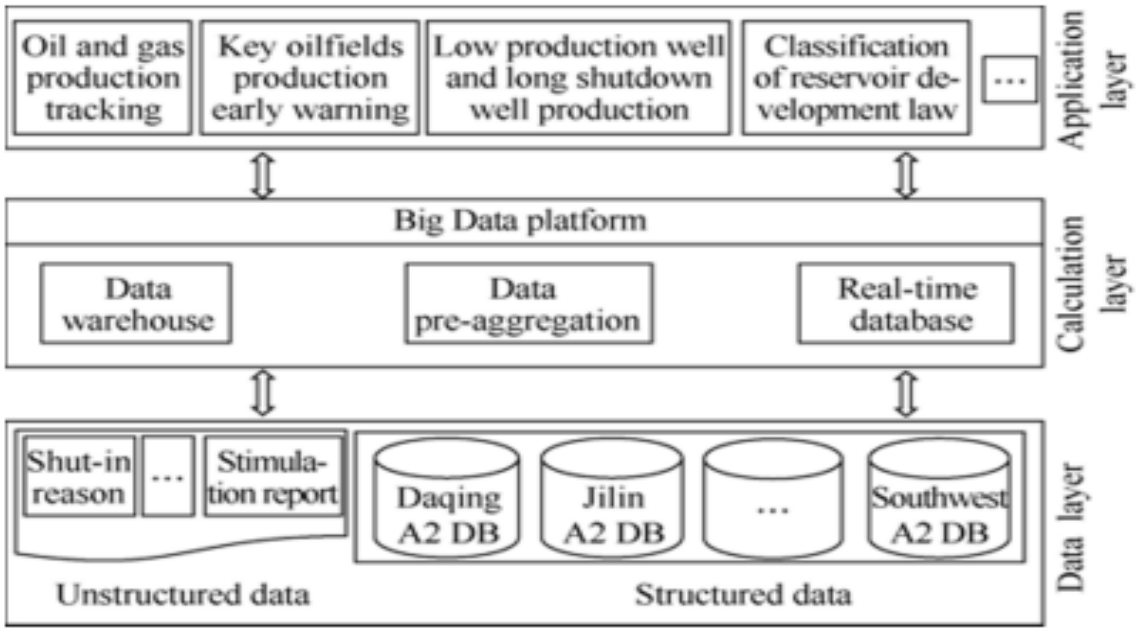
SAFETY AND MAINTAINENCE [(4), pg5, 3.4]:

Apart from the AI application for cost reduction and de-risking, we should mention its extraordinary impact on safety measures. Operations on the oilfields are risky for personnel as there are several risk factors, including heavy equipment, non-covered rotary equipment, high pressure, high-temperature operations, and aggressive chemicals.

BIG DATA

Big data includes unstructured (not organized and text-heavy) and multi-structured data (including different data formats resulting from people/machines interactions). The term Big Data (also called Big Data Analytics or business analytics) defines the first characteristic of this method and that is the size of the available data set. There are other characteristics related to the data which make it viable for Big Data tools. Those characteristics are well named by IBM as three Vs. These three Vs refer to volume, variety, and velocity. However, more recent articles have added two more Vs to give a better definition for Big Data. The additional Vs include veracity and value.

[(3), pg321, 2.1]



[(5), pg1015, fig1.]

BIG DATA IN UPSTREAM OIL AND GAS INDUSTRY

The application of Big Data is now extended beyond the database, marketing, and business techniques. Many engineering disciplines are utilizing Big Data analytics for various applications. Recently, the upstream oil and gas industry is also impacted by the versatility of Big Data. The application of Big Data has become prominent as the amount of data generated and recorded in oil and gas industry has significantly increased. The improvements in seismic acquisitions devices, channel counting, fluid front monitoring geophones, carbon capture and sequestration sites, LWD, and MWD tools have provided vast amount of data to be processed and analysed. [(3), pg323, 3.]

BIG REFING IN BIG DATA

In a study by Plate, the application of Big Data in refining is reviewed. In this case study, the historical data were analysed and processed to improve a petrochemical asset management in a three-step procedure. In this case study, the equipment of interest was a four-stage cracked gas compressor (CGC). The analysis started by first predicting the performance of CGC by analysing the current and historical operating data. In the next phase, based on the device's end-of-life criteria and failure conditions, the performance prediction of the CGC was further tuned. Finally, the estimated performance of the CGC was presented in a user-friendly and visual report to be used for management decisions. These predictive reports which are developed by employing data analysis can significantly reduce the downtime and maintenance costs.[(3), pg325, 4.1]

BIG DATA IN OIL AND GAS TRANSPORTATION

Anagnostopoulos conducted a research to apply Big Data analytics in order to improve the shipping performance. In his study, he aimed to predict the propulsion power to improve the

performances of ships and consequently to lower the greenhouse gas emissions. The data gathered for this study were collected over period of three months from the sensors throughout a LCTC (Large Car Truck Carrier) M/V. In the next step, they used extreme Gradient Boosting (XGBoost) and Multi-Layer Perceptron (MLP) neural networks to conduct the data analysis.[(3), pg325, 4.2]

BIG DATA CHALLENGES

One of the major challenges of Big Data's application in any industry including oil and gas industry is the cost associated with managing the data recording, storage, and analysis. With the recent technological improvements, fog computing, cloud computing, and Internet of Things (IoT) have become available to fix the issues regarding data storage and computations . Costly and limited cloud computing facilities are not suitable options for non-fixed location or latency-sensitive applications. On the other hand, fog computing facilities provide storage and computing facilities closer to data generation sources, which resolves the mentioned challenges to some extent. However, IoT is a newer technology, which is more mobile and fixes the latency issues as well. [(3), pg326, 5.]

In the upstream field, it introduces how to combine digital rock physics with big data and AI to optimize recovery efficiency. Digital core (also known as digital petrophysics—DRP) technology enables more reliable physical information about pore-scale multiphase flows to determine the cause of low recovery and provides new ways for different injection solutions to improve reservoir performance. Combined with digital rock technology and AI, we can integrate the characteristics of digital rock databases into logging and well data, and use a variety of advanced classification techniques to identify the remaining oil and gas potential. Through the multi-phase flow simulation, the multi-scale model can predict the best injection method for maximum recovery under different conditions and propose possible solutions to optimize crude oil production.

In the downstream field, the application of AI and big data analysis in planning and scheduling systems, process unit optimization, preventive maintenance of equipment, and other aspects is introduced. Among them, the molecular-level advanced planning and scheduling system (MAPS) can realize the cost performance measurement under different production schemes for potential types of processable crude oil, which is conducive to more accurate selection of crude oil and prediction of crude oil properties. In addition, the whole process simulation can be used to understand the product quality changes under different crude oil blending schemes and different unit operating conditions, which is conducive to timely adjusting the product blending schemes according to economic benefits or ex-factory demands. The operation conditions of secondary units and even the properties of mixed crude oil can be deduced according to different product quality requirements.[(6), pg181, 1.]

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